

Climate preconditioning and the mechanics of hazardous alpine mass movements

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Abstract

Large rock, ice, and mixed-material collapses initiate process cascades that can travel tens of kilometers or generate locally extreme tsunamis. Although warming alters glacier support, permafrost temperature, meltwater supply, and precipitation phase, the contribution of these changes to individual failures is rarely quantified. Here, I define an event catalog and attribution strategy that separate long-term climate preconditioning from immediate meteorological triggers and from non-climatic controls. The seed catalog resolves several basic ambiguities in recent reports: the Dickson Fjord landslide occurred in 2023 rather than 2024, and the Tracy Arm concern and the large 2025 Alaska landslide describe one event. The catalog also retains the still-active Barry Arm slope as a susceptibility case rather than an occurred collapse. Published mechanisms differ among the cases. The Aru glacier detachments involved basal hydrology and weak till, whereas Chamoli, Dickson Fjord, and Tracy Arm began as rock-slope failures that entrained ice or entered water. These distinctions define observables for an event-centered reanalysis: surface energy balance, liquid precipitation, snow loading, freeze-thaw history, and multidecadal warming. This framework is a first step toward testing systematic climatic association and transferring the resulting mechanical conditions to a global susceptibility assessment.

1 Introduction

Alpine mass movements become especially hazardous when an initial slope failure changes material or enters a confined channel. The 2021 Chamoli rock and ice avalanche, for example, transformed into a debris flow and flood that scoured valley walls and struck two hydropower projects [7]. Fjord failures instead transfer momentum to water. The September 2023 Dickson Fjord rockslide generated a near-source runup of about 200 m and a fjord seiche recorded globally for nine days [5, 9]. The August 2025 Tracy Arm landslide produced a measured maximum runup of 481 ± 7.6 m [8]. The initial failure and the downstream cascade must therefore be described separately.

The mechanical origin of the initial failure remains the central uncertainty. Warming may remove glacier support from a rock slope, thaw ice in fractures, or increase meltwater pressure beneath a glacier. Weather may provide a more immediate perturbation through rain, snow loading, melt, or rapid temperature change. However, lithology, adverse structure, valley geometry, and internal glacier dynamics may be necessary controls. A climatic trend can therefore precondition a failure without being its immediate trigger, and a warm period at failure does not by itself establish causation.

Here, I construct a literature catalog that preserves this distinction and use it to define a test of systematic association with climate. The immediate goal is not to assign an anthropogenic fraction to any one event. It is to identify the atmospheric and cryospheric quantities that each proposed mechanism requires, compare those quantities with reanalysis, and determine whether the events occupy unusual parts of their local climate histories.

2 Catalog construction

The catalog consists of three linked tables. An event table records identity, date, location, setting, initial failure, and downstream transformations. A measurement table stores one quantitative value per row with its units, uncertainty, qualifier, source, and source locator. A claim table records process and attribution statements and labels each as observation, preliminary observation, model output, published inference, or project interpretation. The source key for every row resolves to the accompanying bibliography.

An event is included when rock, glacier ice, or a mixed mass fails rapidly in an alpine or fjord setting and produces exceptional mobility, a damaging cascade, or a substantial prospective tsunami hazard. Snow avalanches without rock or glacier-ice failure and floods without an initial mass movement are outside the present scope. An active slope is retained only when observations define a potential failure volume and a peer-reviewed study connects its motion to a mechanical process. This rule admits Barry Arm but marks it as an active slope rather than an occurred event [2, 6].

The seed catalog emphasizes primary papers and authoritative agency records. The 26 August 2026 Nepal entry is necessarily provisional: the U.S. Geological Survey currently reports a rapid slope failure involving a glacier but cannot yet distinguish collapse of glacier ice from a landslide that incorporated the glacier [10, 11]. No climate attribution is assigned until the source geometry and failure sequence are better constrained.

3 Mechanics represented by the seed events

Table 1 summarizes the current cases. The two Aru events are stored separately because the adjacent glaciers failed two months apart and showed different precursor behavior. Their shared mechanism was unusual: entire low-angle glacier tongues detached after surge-like acceleration. Remote sensing and thermomechanical modeling indicate that increasing melt and rain raised water pressure in weak, fine-grained subglacial sediment. Reduced basal friction transferred resistance toward frozen tongues and lateral margins until those resisting forces were exceeded [3, 4].

The other occurred events began with rock-slope failure. At Chamoli, approximately $26.9 \times 10^6 \text{ m}^3$ of rock and glacier ice detached from Ronti Peak and transformed into a highly mobile debris flow [7]. Geologic structure, steep topography, possible permafrost disturbance, glacier change, and meltwater infiltration are plausible contributors. Snow loading and a preceding temperature anomaly have also been proposed, but no study has isolated a single immediate trigger [12]. At Dickson Fjord, a foliation-parallel rockslide struck a gully glacier before entering the fjord. The reconstructed $25.5 \pm 1.7 \times 10^6 \text{ m}^3$ rock volume and $2.3 \pm 0.15 \times 10^6 \text{ m}^3$ entrained ice volume remain separate catalog measurements because combining them would conceal their different origins [9].

Tracy Arm and Barry Arm illustrate occurred and prospective consequences of glacier retreat along fjords. The minimum mapped Tracy Arm failure volume was $63.5 \pm 2.7 \times 10^6 \text{ m}^3$, and the event was preceded by several days of accelerating microseismicity [8]. The authors infer that climate-driven retreat of South Sawyer Glacier preconditioned the slope; permafrost is unlikely at the site. At Barry Arm, the observed slope displaced 120 m during 2010–2017 while Barry Glacier retreated. The reported correlation between front position and slope displacement is 0.99, supporting glacier debuttressing as a cause of the acceleration [2]. The slope has not failed catastrophically. Modeled tsunami heights depend on the assumed failure geometry; updated scenarios predict about 2.85 m immediately offshore of Whittier in the largest plausible case [1].

Table 1: Seed catalog of focal events and one active slope. Volumes describe the initial failed mass. Statements in the last column preserve the claim strength of the cited source.

Event and date	Initial failure	Volume (10^6 m^3)	Cascade	Climate evidence
Nepal, 26 Aug 2026	Glacier-involving slope failure	unresolved	Debris flow and flood traveled about 100 km	No event-specific attribution; source classification remains provisional [11]
Chamoli, 7 Feb 2021	Rock and glacier ice	27	Avalanche to debris flow and flood	Several thermal effects are plausible; no single trigger isolated [7]
Aru-1, 17 Jul 2016	Glacier detachment	68 ± 2	Low-angle ice avalanche, runout up to 7 km	Multiyear melt and rain inferred to increase subglacial water pressure [3]
Aru-2, 21 Sep 2016	Glacier detachment	83 ± 2	Low-angle ice avalanche, runout about 5 km	Same forcing and substrate class as Aru-1 [3, 4]
Dickson Fjord, 16 Sep 2023	Foliation-parallel rockslide	25.5 ± 1.7 rock	Glacier entrainment, tsunami, and nine-day seiche	Glacier thinning inferred to debutstress the toe; proximate trigger unresolved [9]
Tracy Arm, 10 Aug 2025	Bedrock landslide	63.5 ± 2.7 minimum mapped	Glacier/fjord impact, tsunami, and 36-hour seiche	Anthropogenic warming linked quantitatively to glacier-retreat preconditioning [8]
Barry Arm, active	Slow bedrock landslide	455	Potential rapid failure and tsunami	Acceleration strongly correlated with glacier retreat [2]

4 Event-centered climate test

The first analysis will test conditions required by the cataloged mechanisms. For each event, atmospheric reanalysis will provide near-surface air temperature, precipitation phase and amount, snow accumulation, radiative and turbulent fluxes, and surface pressure. These fields will be reduced to five mechanically interpretable predictors: cumulative positive degree days, liquid-water input, recent snow load, freeze–thaw crossings, and a multidecadal temperature anomaly. Windows will be defined before examining event values so that the analysis does not select a favorable timescale after the fact.

The primary comparison is local and seasonal. Each event value will be ranked against the same calendar window at that location over the full reanalysis period. This removes much of the mean climatic contrast between Tibet, Greenland, Alaska, and the Himalaya. A second comparison will use

matched non-event years at each site. The null hypothesis is that event conditions are exchangeable with these local controls. Rejection for one predictor would establish an association, not a complete causal attribution.

Global products smooth the elevation and aspect that govern energy balance in steep terrain. Downscaling will therefore begin with the smallest correction that addresses a named physical bias: temperature adjusted between grid-cell and source elevation using a documented lapse rate, precipitation separated by phase, and shortwave radiation corrected for slope and aspect. Results will be reported both with and without each correction. More complex regional modeling is warranted only if these corrections change the sign or rank of an event anomaly.

5 Uncertainty and tests of interpretation

The strongest current climate statements concern preconditioning rather than immediate triggers. At Dickson Fjord, geometry and multidecadal glacier thinning support debuttressing, but the published study is not a formal event attribution and does not identify unusual weather on 16 September 2023 [9]. At Barry Arm, slope motion and glacier retreat covary, but a possible precipitation response was described as tenuous [2]. These examples require different tests: a multidecadal glacier or temperature history for preconditioning, and hourly-to-seasonal atmospheric fields for an immediate trigger.

Catalog incompleteness is a second source of uncertainty. The present cases were selected because they motivated the study, not by a systematic global search. They cannot support a frequency trend. The next catalog version must define databases, search terms, language coverage, minimum documented size, and treatment of events discovered only through seismic or satellite records. Negative controls are equally important because detection probability has increased with modern sensors.

6 Conclusions

The seed catalog establishes a traceable distinction among initial failure, downstream cascade, climate preconditioning, and immediate trigger. Its central result is methodological: recent events that are often grouped together represent different mechanical systems, and their climate hypotheses require different atmospheric tests. Aru directly motivates a basal-water analysis; Dickson Fjord, Tracy Arm, and Barry Arm motivate glacier-retreat and debuttressing histories; Chamoli and the provisional Nepal event require greater caution about source classification. Event-centered reanalysis can test whether the proposed atmospheric conditions were unusual. A systematic catalog and matched non-events will then be needed before these relationships can be transferred to a global hazard assessment.

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